

Notes Optimal transit – combining models to create an improved bus system

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1: Model 1, Monge-Kantorovich Problem

From [1]

1.1 Monge

The original Monge problem was as follows. Given a countryside with mines and factories how can we optimally match up each factory with each mine so that every factory is getting the ore it needs and the mines are getting rid of all of their ore.

The factories demand is in one space M^- and the mines supply is in another M^+ . The text gets into detail about the specifics of what is required of these spaces. i.e. Metric, separable, Complete, and geodesic but for anything I am likely to be doing it is sufficient to know that R^2 satisfies all of these requirements.

1. μ^+ and μ^- are Borel Probability measures. They take a set in M^+ and M^- , respectively, and tell us how much supply or demand there is in that set
2. $G\#$ is a function that takes points in μ^+ and moves them to μ^- . It 'pushes forward' the points from M^+ . In a way it is telling us how much demand is being 'sent' to each mine

$$(G\#\mu^+)[v] := \mu^+[G^{-1}(v)] \forall V \subset M^- \quad (1)$$

This equation tells us that the demand over a region v in M^- must be equal to the amount supplied to that region given the push forward G

$$\text{cost} = \int_{\mu^+} c(x, G(x)) d\mu^+(x) \quad (2)$$

This equation gives us the cost function we want to minimize. It sums all points in μ^+ and finds the cost of sending each point to where it is going using cost function c more on this in the next paragraph. The goal of this optimization is to find the G that minimizes this cost

The first cost function that Monge tried was $c(x, y) = |x - y|$ This creates problems when trying to solve this and $c(x, y) = (x - y)^2$ has much better properties. I don't fully understand why but it connects to the Hilbert space geometry and according to Brenier's Theorem using this cost function means there is only one minimizing G and the solution G is a convex gradient i.e. $G = \nabla\phi$ where ϕ is convex

$$W_c(\mu^+, \mu^-) = \inf \text{cost}(G) \quad (3)$$

This is the minimum cost to perfectly match μ^+ with μ^- we can think of it of how different or how 'far apart' two distributions are and we call it a Wasserstein Distance

CONTINUE THE REST OF THIS MONGE STUFF EVEN IF BRIEFLY

1.2 Kantorovich

Kantorovich relaxed the problem by allowing mines to split up where ore is sent to multiple factories. This relaxation makes the problem much easier to solve and more realistic

$$\text{cost}(\gamma) = \int_{M^+ \times M^-} \text{cost}(x, y) d\gamma(x, y) \quad (4)$$

$M^+ \times M^-$ is the cross product meaning we are looking at every pairing of points across M^+ and M^- each one of these pairings gets a value of how much ore is sent between these points which is not possible in the Monge version of the problem. This reworking makes our cost function linear since γ is outside of the cost function, this makes it an easier problem to deal with.

We can think about this problem as a matrix where each row is a point in M^+ and each column in this row represents a point on M^- the values in each spot represent the amount of ore that will go from that point on M^+ to the corresponding one on M^- . This is exactly what happens when we approximate it in code with discrete node locations.

$\Gamma(\mu^+, \mu^-)$ is the set of all γ that map out where the ore goes. We know this is a convex set meaning any two γ have a valid line between them, this was not the case for G. We also know that a solution to this problem does exist but uniqueness and characterization are still unknown.

2: Model 2: Mixed Integer Linear Programming: Joint Optimization Problem

From [2]

2.1 Vocab and notation

1. headway- the interval between two trains traveling the same path. For this model the headways are pre chosen as discrete options from a list. For example options=[1,5,10,15]
2. O-D- a pair of origin and destination
3. perceived headway- the wait time felt by passengers waiting in a platform. This accounts for all train patterns that are running through that stop
4. pattern- this refers to one way a train can travel from an origin to a destination, in other words this is what stops a given train actually stops at and what stops it skip when going from O to D
5. route- In this paper this refers to a section of track that trains need to travel on, for example this would be the section of rail running north from downtown but not the purple or red line, these would be patterns.
6. M- Big M, this is used as a place holder for a sufficiently big number. In reality this should be several different big M's for each of the numbers needed (like number of people that can fit on a train)

2.2 Pre-computing

Before they begin optimizing this paper pre computes some data. They compute perceived headway given different combinations of patterns and headway for each pattern. For a simple version with 2 patterns and 2 headway option would yield the combination set $[(0,1),(0,2),(1,0),(1,1),(1,2),(2,0),(2,1),(2,2)]$ each index is a pattern that the train could take and the number in that index is the headway choice that that pattern runs with. 0 means that this pattern is not in service, we are assuming that at least one of the patterns must be running at all times. For example (0,1) would mean pattern one is not running and pattern 2 is running with the first option in the headway list. This list of all combinations for a route is called C_r . The perceived headway is computed assuming all patterns stop at the stop we are calculating the perceived headway for. The patterns here are not actuality known yet and we don't need this for the calculation. The number of patterns that are considered is an input and the optimization can always make patterns inactive by setting them to zero. Below is the equation used to calculate perceived headway, this is done by converting time to frequency, summing frequency and converting back to time.

$$T_c^\omega = \left(\sum_{p \in \mathcal{P}_r | h_{cp}^\gamma \neq 0} \left(T_{h_{cp}^\gamma}^\gamma \right)^{-1} \right)^{-1}, \forall c \in \mathcal{C}_r \quad (5)$$

2.3 Binary variables

There are four types of binary variable and they are listed below. I will explain each one in order

1. $x_{pijh}^\eta \in \{0, 1\}$
2. $x_{pij} \in \{0, 1\}$
3. $y_{ph} \in \{0, 1\}$
4. $z_{i,d,c} \in \{0, 1\}$

x_{pijh}^η and x_{pij} are similar types of variable and could be condensed into one but are not for computing and efficiency purposes. x_{pij} determines the pattern arc (stop sequence) that a pattern uses. This is like a map telling one pattern what stops it stops at and what stops it skips. If $x_{pij} = 1$ then pattern p travel directly from stop i to stop j. If this variable is a zero then this is not the case. x_{pijh}^η is very similar but with the included parameter of headway. These two variables are connected and are separated into two for efficiency.

y_{ph} is headway assignment for each pattern, each pattern is assigned one headway from the list and for that pattern and headway it gets a 1. Otherwise this variable will be zero

$z_{i,d,c} \in \{0, 1\}$ is a binary variable. Looking at every O-D pair on a route (i=origin, d=destination) every $c \in \mathcal{C}_r$ gets a z value. If every pattern in c stops at station i and station d that $z=1$ otherwise it equals zero. Essentially they use this in combination with C_r to turn the problem into an easier problem to optimize so they don't need to deal with finding perceived head ways and dividing they can instead just choose from this selection menu

2.4 Continuous variables

1. Entry flow

$$\begin{aligned} f_{dic}^\omega &\geq 0, \quad i \notin \{d, \tilde{d}\} \\ f_{dic}^\omega &= 0, \quad i \in \{d, \tilde{d}\}, \quad \forall d \in \mathcal{D}_r, i \in \mathcal{S}_r, c \in \mathcal{C}_r \end{aligned}$$

2. Boarding flow

$$\begin{aligned} f_{dicp}^\alpha &\geq 0, \quad i \notin \{d, \tilde{d}\}, h_{cp}^\gamma > 0 \\ f_{dicp}^\alpha &= 0, \quad i \in \{d, \tilde{d}\} \text{ or } h_{cp}^\gamma = 0, \quad \forall d \in \mathcal{D}_r, i \in \mathcal{S}_r, p \in \mathcal{P}_r, c \in \mathcal{C}_r \end{aligned}$$

3. Inter-stop flow

$$\begin{aligned} f_{dpij}^\lambda &\geq 0, \quad i \notin \{d, \tilde{d}\}, i < j \\ f_{dpij}^\lambda &= 0, \quad i \in \{d, \tilde{d}\} \text{ or } i \geq j, \quad \forall d \in \mathcal{D}_r, i, j \in \mathcal{S}_r, p \in \mathcal{P}_r \end{aligned}$$

4. Exit flow

$$f_{jp}^\beta \geq 0, \quad \forall j \in \mathcal{S}_r, p \in \mathcal{P}_r$$

These are the flow variables which are also being optimized. They describe how many people are doing an action in one time interval. They respectively tell us 1- The number of people entering a station i who are going to station d with perceived headway c , 2- the number of people boarding a train with pattern p from stop i , going to stop d , with combo c , 3- the number of people traveling from stop i to stop j on pattern p who are eventually headed to stop d , 4- the number of people leaving a train traveling pattern p at stop j

2.5 Constraints

1. Constraint 1 (Eq. 15): Pattern Flow Balance

$$\sum_{i \in \mathcal{S}_r} x_{pij} = \sum_{k \in \mathcal{S}_r} x_{pjk}, \quad \forall j \in \mathcal{S}_r, p \in \mathcal{P}_r$$

Ensures the number of incoming links equals the number of outgoing links at every stop, forming a closed loop for each pattern.

2. Constraint 2 (Eq. 16): At Most One Incoming Link

$$0 \leq \sum_{i \in \mathcal{S}_r} x_{pij} \leq 1, \quad \forall j \in \mathcal{S}_r, p \in \mathcal{P}_r$$

Restricts each stop to have at most one incoming link, preventing branching.

3. Constraint 3 (Eq. 17): Big-M Flow Activation

$$f_{dpij}^\lambda \leq M x_{pij}, \quad i < j, \quad \forall d \in \mathcal{D}_r, p \in \mathcal{P}_r, i, j \in \mathcal{S}_r$$

Flows from i to j can only be as big as M but will be zero if that link is 'turned off' by x

4. Constraint 4 (Eq. 19): One Headway Per Pattern

$$\sum_{h \in \mathcal{H}_r} y_{ph} = 1, \quad \forall p \in \mathcal{P}_r$$

Assigns exactly one headway to each pattern.

5. **Constraint 5 (Eq. 21): Headway-Stop Consistency**

$$x_{pijh}^\eta \leq y_{ph}, \quad i \neq j, \quad \forall i, j \in \mathcal{S}_r, \quad p \in \mathcal{P}_r, \quad h \in \mathcal{H}_r$$

Makes headway for x^η match the headway of it's pattern .

6. **Constraint 6 (Eq. 22): Headway Aggregation**

$$\sum_{h \in \mathcal{H}_r} x_{pijh}^\eta = x_{pij}, \quad i \neq j, \quad \forall i, j \in \mathcal{S}_r, \quad p \in \mathcal{P}_r$$

Forces every active link to have exactly one headway assigned.

7. **Constraint 7 (Eq. 23): Capacity Constraint**

$$\sum_{d \in \mathcal{D}_r} f_{dpij}^\lambda \leq B_r \sum_{h \in \mathcal{H}_r} \frac{1}{T_h^\gamma} x_{pijh}^\eta, \quad \forall i, j \in \mathcal{S}_r, \quad p \in \mathcal{P}_r$$

Limits total passengers on a link to the capacity provided by trains running on it.

8. **Constraint 8 (Eq. 24): Fleet Size**

$$\sum_{p \in \mathcal{P}_r} \sum_{i \in \mathcal{S}_r} \sum_{j \in \mathcal{S}_r} T_{ij}^\lambda \sum_{h \in \mathcal{H}_r} \frac{1}{T_h^\gamma} x_{pijh}^\eta \leq N_r^\nu$$

Ensures the total fleet required across all patterns does not exceed the available fleet.

9. **Constraint 9 (Eq. 25): One Combination Per O-D Pair**

$$\sum_{c \in \mathcal{C}_r} z_{idc} \leq 1, \quad i \notin \{d, \tilde{d}\}, \quad \forall d \in \mathcal{D}_r, \quad i \in \mathcal{S}_r$$

Allows each O-D pair to be assigned at most one pattern-headway combination.

10. **Constraint 10 (Eq. 26): Combination-Headway Consistency**

$$z_{idc} \leq y_{ph}, \quad i \notin \{d, \tilde{d}\}, \quad h = h_{cp}^\gamma \neq 0, \quad \forall i \in \mathcal{S}_r, \quad d \in \mathcal{D}_r, \quad c \in \mathcal{C}_r, \quad p \in \mathcal{P}_r$$

Ensures a combination is only assigned if its headways match those assigned to the patterns.

11. **Constraint 11 (Eq. 27): Flow Bound by Combination**

$$f_{dicp}^\alpha \leq M z_{idc}, \quad i \notin \{d, \tilde{d}\}, \quad \forall d \in \mathcal{D}_r, \quad i \in \mathcal{S}_r, \quad c \in \mathcal{C}_r, \quad p \in \mathcal{P}_r$$

Doesn't let boarding happen if that pattern isn't in their combination(the train won't get they where they need to go).

12. **Constraint 12 (Eq. 28): Frequency Share Rule**

$$-M(1 - z_{idc}) \leq T_{h_1}^\gamma f_{dicp_1}^\alpha - T_{h_2}^\gamma f_{dicp_2}^\alpha \leq M(1 - z_{idc})$$

Distributes flow among patterns in proportion to their frequencies when a combination is assigned.

13. **Constraint 13 (Eq. 29): Entry Flow Equals Demand**

$$\sum_{c \in \mathcal{C}_r} f_{doc}^\omega + \sum_{c \in \mathcal{C}_r} f_{d\tilde{o}c}^\omega = E_{od}, \quad \forall o, d \in \mathcal{D}_r$$

Ensures the total entry flow at both platforms of an origin equals the demand.

14. **Constraint 14 (Eq. 30): Exit Flow Equals Demand**

$$\sum_{p \in \mathcal{P}_r} f_{dp}^\beta + \sum_{p \in \mathcal{P}_r} f_{\tilde{d}p}^\beta = \sum_{o \in \mathcal{D}_r} E_{od}, \quad \forall d \in \mathcal{D}_r$$

Ensures the total exit flow at both platforms of a destination equals total demand.

15. **Constraint 15 (Eq. 31): Entry Equals Boarding**

$$f_{dic}^\omega = \sum_{p \in \mathcal{P}_r} f_{dicp}^\alpha, \quad i \notin \{d, \tilde{d}\}, \quad \forall d, i \in \mathcal{S}_r, \quad c \in \mathcal{C}_r$$

Ensures every passenger who enters the system boards a pattern.

16. **Constraint 16 (Eq. 32): Flow Balance at Intermediate Stops**

$$\sum_{c \in \mathcal{C}_r} f_{djc}^\alpha + \sum_{i=0}^{j-1} f_{dpij}^\lambda = \sum_{k=j+1}^{|\mathcal{S}_r|} f_{dpjk}^\lambda, \quad j \notin \{d, \tilde{d}\}$$

Ensures boardings plus arrivals equal continuations at non-destination stops.

17. **Constraint 17 (Eq. 33): Arriving Equals Exiting**

$$\sum_{i=0}^{j-1} f_{dpij}^\lambda = f_{jp}^\beta, \quad j \in \{d, \tilde{d}\}, \quad \forall d \in \mathcal{D}_r, \quad p \in \mathcal{P}_r$$

Ensures passengers arriving at their destination stop exit the system.

18. **Added constraint, not in the paper: Assures that if a stop is skipped it cannot have any trains coming in or out. (this has to be only applied to the forward direction to allow for the model to 'complete' the loop)**

$$\sum_{k,l \in \mathcal{S}_r} x_{pkl} = 0 \quad \text{if } x_{pij} = 1 \text{ where } i > k > j, \text{ or } i > l > j$$

2.6 Optimizing

$$\begin{aligned} \min_{x_{pij}, x_{pijh}^\eta, y_{ph}, z_{ide}, f_{dic}^\omega, f_{dicp}^\alpha, f_{dpij}^\lambda, f_{jp}^\beta, f_{diipc}^\chi} & \sum_{d \in \mathcal{D}_r} \sum_{p \in \mathcal{P}_r} \sum_{i \in \mathcal{S}_r} \sum_{j=i+1}^{|\mathcal{S}_r|} T_{ij}^\lambda f_{dpij}^\lambda \\ & + \gamma^\omega \sum_{d \in \mathcal{D}_r} \sum_{i \in \mathcal{S}_r} \sum_{c \in \mathcal{C}_r} \frac{T_c^\omega}{2} f_{dic}^\omega \\ & + \gamma^\chi \sum_{d \in \mathcal{D}_r} \sum_{i \in \mathcal{S}_r} \sum_{j \in \{i, \tilde{i}\}} \sum_{p \in \mathcal{P}_r} \sum_{c \in \mathcal{C}_r} \left(\frac{T_c^\omega}{2} + T^\chi \right) f_{diipc}^\chi \end{aligned} \quad (14)$$

This is used as the cost function that is being minimized and both the binary and continuous variables are used to minimize this cost.

$$C_r(n_r) = \text{minimum passenger cost for a route given } n_r \text{ vehicles assigned to this route} \quad (6)$$

The sum of this c_r for each route is then minimized based on the n_r number of vehicles assigned to a given route.

The MILP takes the cost function c , a vector of 1's and telling it witch variables used are continuous and witch ones are discreet, a l and u vector telling it what the upper and lower bounds of each variable are (for binary this is 1 and 0), and the constraints. The constraints consist of a matrix A witch gives the coefficients for the variables that are included in that constraint and then two vectors b_l and b_u with as many entries as there are constraints that give the lower and upper bounds of the constraints. If it is an equation b_l and b_u will be the same number but if it's an inequality they can be different.

The MILP works by first solving the problem assuming everything is continuous. If the solution already uses integers than they are done. If not they turn the problem into multiple sub problems or 'branches' then go through and resolve or cut these to trim the problem down until they have the right answer.

They use lower and upper bounds and trim the branch when theses don't match. !!!!!!!!!!!!!Understnd this better

3: Model 3, Zonal connections

From [3]

3.1 Introduction and Vocab

This paper aims to find an optimal city wide transport plan using multiple modes of transit to get people effectively where they need to go. It divides up the city into 'zones' and tells us what infrastructure should be built and what connections should be available between zones as well as how many people use these connections all of witch is done to minimize total door to door travel time

1. access and egress - the part of your journey that is getting from the origin to the 'bus stop' or equivalent and the part getting from the 'bus stop' to your destination respectively
2. feeder services - stuff that helps with access and aggress portions of trip, think a shuttle that takes you too and from the station
3. clustering - This is combining similar trips into one cluster of trips i.e. if two people have similar origins and destinations this can be made into one cluster. This is essential since we are dealing with city wide problems here.
4. Corridors - Major travel routes, think North South on the lake in Chicago
5. Centroid - This is the 'center' of each zone. It is not the geometrical center but the most important part that the zone was built around to minimize distances
- 6.

The first step of the process is placing the centroids around the city. This is done by minimizing the below equation where N_a is amount of demand at point P_a in the city (these are given as data already) and μ_i is the location of the centroids. This is minimized to get everyone as close as they can be to a centroid.

$$\min_{\mu_i} \sum_{i=1}^{|Z|} \sum_{a \in \mathcal{A}_i} (N_a^t)^2 \|P_a - \mu_i\|_2^2$$

The next step is creating the zones which they doing using a 'Voronoi diagram' but all this means is each point is assigned to the closest geographical centroid

They also deal with clustering in this stage, they take into account angle between O-D pairs when determining if there should be a cluster since this means people are traveling in the same direction or not

The last thing they do before beginning to optimize is determine $(i, j) \in L_m$ which is the list of all possible links. Because this is such a big problem they do not include every possible link, instead they narrow it down based on proximity, demand, and Pre-existing connections

3.2 Zonal connection Variables

There are two types of integer variable in this model and they are listed below. I will explain each one in order.

1. $x_{mij}^\lambda \in \mathbb{Z}_{\geq 0}$

2. $x_{mij}^\omega \in \mathbb{Z}_{\geq 0}$

x_{mij}^λ is the number of connections between zones i and j of type m that are actively used. This is not a binary variable because for example there could be 3 bus routes between zone 1 and 2
 x_{mij}^ω is the number of new connections that must be built between zone i and j given mode m. If they already exist this does not effect this variable and for modes that don't need infrastructure built like walking this is always zero

1. Starting flow

$$f_{om}^\alpha \geq 0, \quad \forall o \in \mathcal{Z}, m \in \mathcal{M}$$

2. Ending flow

$$\begin{aligned} f_{omd}^\beta &\geq 0, \quad d \neq o \\ f_{omd}^\beta &= 0, \quad d = o, \quad \forall o, d \in \mathcal{Z}, m \in \mathcal{M} \end{aligned}$$

3. Interzonal flow

$$\begin{aligned} f_{omij}^\lambda &\geq 0, \quad o \neq j \text{ and } (i, j) \in \mathcal{L}_m \\ f_{omij}^\lambda &= 0, \quad o = j \text{ or } (i, j) \notin \mathcal{L}_m, \quad \forall o, i, j \in \mathcal{Z}, m \in \mathcal{M} \end{aligned}$$

4. Intermodal transfer flow

$$\begin{aligned} f_{om_1m_2i}^\chi &\geq 0, \quad m_1 \neq m_2 \\ f_{om_1m_2i}^\chi &= 0, \quad m_1 = m_2, \quad \forall o, i \in \mathcal{Z}, m_1, m_2 \in \mathcal{M} \end{aligned}$$

These represent respectively, the flow of people starting in zone o using mode m, the flow of people ending in zone d with origin zone o using mode d, the flow from zone i to zone j with mode m originating in zone o, and the flow of people originating in zone o who transfer from mode m_1 to mode m_2 in zone i.

3.3 Optimizing zonal connections

The first optimization step is to figure out what zones should be connected by what modes, this may involve building infrastructure or just setting bus lines. We set these variables to be constrained so that the output of the optimization is feasible and useful, The constraints given are equations (13)-(29) in the document

We optimize these variables based on the following cost function which can be broken down into 4 parts

1. Starting costs- The time taken to walk to a centroid and wait for a mode of transit
2. inter zonal costs- time taken to get from zone to zone using any mode
3. transfer costs- time taken to exit and re-enter differing modes of transit
4. ending costs- getting from centroid to final destination

$$\min_{\substack{f_{om}^\alpha, f_{omd}^\beta, f_{omij}^\lambda, f_{om_1m_2}^\chi \\ x_{mij}^\lambda, x_{mij}^\omega}} \sum_{o \in \mathcal{I}} \sum_{m \in \mathcal{M}} \left[\gamma^r \left(f_{om}^\alpha c_{mo}^\alpha + \sum_{(i,j) \in \mathcal{L}_m} f_{omij}^\lambda c_{mij}^\lambda + \sum_{m_1 \in \mathcal{M}} \sum_{i \in \mathcal{I}} f_{omm_1i}^\chi c_{mm_1i}^\chi + \sum_{d \in \mathcal{I}} f_{omd}^\beta c_{md}^\beta + \sum_{(i,j) \in \mathcal{L}_m} \frac{f_{omij}^\alpha}{R_m^d} \right) \right. \\ \left. + \left(f_{om}^\alpha + \sum_{i \in \mathcal{I}} f_{imo}^\beta \right) \frac{p_m^o}{R_m^f} (\gamma_m^{of} t_{mo}^f + \gamma_m^{ef} d_{mo}^f) \right]$$

3.4 Route Generation – MILP Decision Variables

Continuous decision variables

- f_{oijmu}^ν : Interzonal flow, originating from zone $o \in \mathcal{Z}$, which travels on route segment $u \in \mathcal{U}_{mij}$ of transit mode $m \in \mathcal{M}_t$ from zone $i \in \mathcal{Z}$ to zone $j \in \mathcal{Z}$.
- f_{omiu}^ψ : Starting flow entering route segment $u \in \mathcal{U}_{mij}$, $j \in \mathcal{Z} \mid (i, j) \in \mathcal{L}_m^\lambda$ of transit mode $m \in \mathcal{M}_t$, originating from zone $o \in \mathcal{Z}$.
- f_{omju}^η : Ending flow leaving route segment $u \in \mathcal{U}_{mij}$, $i \in \mathcal{Z} \mid (i, j) \in \mathcal{L}_m^\lambda$ of transit mode $m \in \mathcal{M}_t$, originating from zone $o \in \mathcal{Z}$.
- $f_{omju_1u_2}^\delta$: Direct route flow between route segment $u_1 \in \mathcal{U}_{mij}$ which is directly connected to $u_2 \in \mathcal{U}_{mjk}$ in zone j of transit mode $m \in \mathcal{M}_t$, where $i, k \in \mathcal{Z} \mid (i, j), (j, k) \in \mathcal{L}_m^\lambda$.
- $f_{omju_1u_2}^\tau$: Transfer flow between route segment $u_1 \in \mathcal{U}_{mij}$ which is not directly connected to $u_2 \in \mathcal{U}_{mjk}$ in zone j of transit mode $m \in \mathcal{M}_t$, where $i, k \in \mathcal{Z} \mid (i, j), (j, k) \in \mathcal{L}_m^\lambda$.

Binary decision variables

- $x_{mju_1u_2}^\delta$: Binary route segment connection variable that $x_{mju_1u_2}^\delta = 1$ if the route segment $u_1 \in \mathcal{U}_{mij}$ is directly connected to $u_2 \in \mathcal{U}_{mjk}$ in zone $j \in \mathcal{Z}$ for transit mode $m \in \mathcal{M}_t$, where $i, k \in \mathcal{Z} \mid (i, j), (j, k) \in \mathcal{L}_m^\lambda$; $x_{mju_1u_2}^\delta = 0$ otherwise.

3.5 Route Optimization

Now that we have the zonal connections we optimize the actual routes that should be traveled. We set constraints on the above variables in equations (33)-(44) and minimize them based on the following cost function. We are aiming to minimize transfers here and that means specifically multi modal transfers. We assume that within one mode of transport there are no transfer costs

$$\min_{\substack{x_{mju_1u_2}^\delta, f_{omiu}^\psi, f_{omju}^\eta, \\ f_{oijmu}^\nu, f_{omju_1u_2}^\delta, f_{omju_1u_2}^\tau}} \sum_{(i,j),(j,k) \in \mathcal{L}_m^\lambda} \sum_{u_1 \in \mathcal{U}_{mij}} \sum_{\substack{u_2 \in \mathcal{U}_{mjk} \\ o \in \mathcal{Z}}} f_{omju_1u_2}^\tau, \quad \forall m \in \mathcal{M}^t$$

3.6 Self notes

I don't love the two step optimization but it makes the problem much easier to solve and the time scales are different and it probably doesn't miss anything if done right

4: Model 4: Optimizing elevators using MILP

From [4]

This model solves a MILP problem every time a new elevator call is made. Every action needed to be made by an elevator is a 'node' and each elevator travels on 'arcs' from node to node completing all the tasks that need to be done. These arcs are constrained to be feasible and assigned a cost for each completed arc. These arcs are then minimized based on average waiting time for the elevator not total travel time. I think this is not very applicable since it is re-calculated every time a call is made the variables are hard to integrate

$$\text{minimize } \left\{ \sum_{i \in P_1} \omega_i t_i / \sum_{i \in P_1} \omega_i \right\} \quad (1)$$

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